

EDGE MINDS HACKATHON 2026

Team : Electro_Neurons (ID : EDGM26-DWDPMR)

Project Title : Autonomous Local Research Agent Using LLaMA 3.2 (1B)

Introduction

Research tasks often require more than a single prompt. A useful research system must break a topic into smaller questions, search for relevant information, filter useful evidence, summarize findings, and compile them into a structured final report.

This project implements an autonomous local research agent powered by the LLaMA 3.2 (1B parameter) Small Language Model, deployed on the NVIDIA Jetson Orin Nano. The model acts as the central reasoning engine, planning sub-questions, searching local text files, generating evidence-backed summaries, and producing a final report without intermediate user guidance.

This approach ensures low-latency offline operation, improved auditability, and strong privacy, making it suitable for environments where external APIs cannot be used.

Objectives

Design a lightweight AI research assistant using LLaMA 3.2 (1B) as the core reasoning model.

Enable the system to autonomously generate sub-questions and decide execution flow (agentic behavior).

Implement local document search over a folder of text files without internet dependency.

Extract and summarize relevant evidence for each sub-question.

Compile structured, coherent final reports from intermediate summaries.

Optimize the system for efficient deployment on Jetson Orin Nano.

Methodology

The system follows a multi-stage agentic pipeline driven by LLaMA 3.2 (1B):

User Input

The user provides a research topic.

Planning (SLM - LLaMA 3.2)

The model generates 3-6 sub-questions and determines the order of execution.

Local Retrieval Tool

A file search module scans local text documents, extracts relevant snippets, and returns file references.

Reasoning + Summarization (SLM)

LLaMA 3.2 processes retrieved content to generate concise, evidence-based answers for each sub-question.

Iterative Agent Loop

The model evaluates coverage and decides whether additional sub-questions are needed.

Report Compilation

All summaries are merged into a structured final research report.

The architecture is optimized for edge deployment, ensuring low memory footprint and efficient inference using the 1B parameter model.

Scope

Offline academic research assistant

Local knowledge base querying system

Privacy-sensitive environments (defense, healthcare, enterprise)

Edge AI deployment on Jetson devices

Future improvements may include:

Document chunking and embedding-based retrieval

Vector search integration

Confidence scoring for retrieved evidence

Persistent memory and improved agent state tracking

Fine-tuning or prompt optimization specific to LLaMA 3.2 (1B)